

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0702

COURSE OUTCOME COVERED

CO5:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL5)

Assessment Tool Weightage: 40%

QUESTIONS: 3

Total Marks:30

Annexure A

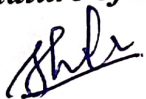
Constraint-Based Problem Solving

Code of Conduct:

192511445

I SHAIKHAJAH RUSHAN (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

Constraint-Based Problem Solving

1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability.
2. A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.
3. A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Problem & Constraints	20%	Clearly identifies all constraints and fully understands the problem requirements	Identifies most constraints with minor gaps in understanding	Identifies some constraints but misses key aspects	Shows little or no understanding of constraints or problem
Problem Decomposition	15%	Breaks problem into logical, well-structured sub-problems effectively	Breaks problem into sub-parts with some logical flow	Attempts decomposition but lacks clarity or structure	No clear decomposition of the problem
Application of Constraints in Solution	20%	Applies all constraints correctly and consistently in solution design	Applies most constraints correctly with minor errors	Applies some constraints but with noticeable errors	Fails to apply constraints appropriately
Logical Reasoning & Strategy	15%	Uses strong, clear, and efficient reasoning strategies	Uses generally sound reasoning with minor gaps	Reasoning is partially correct but inconsistent	Reasoning is unclear or incorrect
Solution Accuracy & Feasibility	15%	Solution is fully correct, feasible, and optimized	Solution is mostly correct and feasible	Solution has errors or is partially feasible	Solution is incorrect or not feasible
Creativity & Innovation	5%	Demonstrates highly innovative and unique approach	Shows some creativity in approach	Limited creativity shown	No creativity; relies on basic or copied ideas
Clarity of Presentation	10%	Solution is clearly presented with excellent organization and explanation	Presentation is clear with minor issues	Presentation lacks clarity or organization	Poorly presented and difficult to understand

Question 1 :

Rural Branch Office - Internet Connection,
Devices and Security.

Problem Analysis

The multinational company is opening a branch office in a remote rural area. The main constraints are:

- High-speed wired Internet is unavailable.
- The office requires Internet access for cloud applications.
- Secure communication with headquarters is essential.
- The budget is limited.
- Business operations should continue even if the primary connection fails.
- The solution should support future expansion.

Recommended Internet connection

I recommend using 4G/5G wireless broadband as the primary Internet connection, if mobile coverage is available. If reliable cellular coverage is unavailable, Satellite Internet can be used as the primary

connection.

For uninterrupted operation, a dual-WAN connection should be implemented. For example:

- Primary connection: 5G/4G broadband.
- Automatic failover switches traffic to the backup connection when the primary link fails.

This provides better reliability without requiring expensive wired infrastructure.

Recommended Network Devices

1. Dual-WAN Router

- Connects multiple Internet links.
- Automatically performs failover.
- Supports load balancing where appropriate.
- Provides VPN functionality.

2. Managed Ethernet Switch

- Connects computers, printers, servers, IP phones and other wired devices.
- VLAN support allows the network to be divided into different logical segments.

3, Wireless Access Points

- Provide Wi-Fi connectivity to employees.
- Business-grade access points should be selected for better security and management.

4, Firewall /UTM Device

- Filters unwanted traffic.
- Provides VPN, intrusion prevention and application control.
- Protects the branch network from Internet-based attacks

Security Mechanisms

Secure communication with headquarters should be established using a site-to-site IP sec VPN.

Other security measures include:

- WPA3/WPA2 - enterprise for wireless security
- Strong user authentication.
- Firewall rules.
- Intrusion detection/prevention.

- Endpoint antivirus and Security Software
- Multi-factor authentication for important systems.

Bandwidth considerations

The cloud applications and video conferencing can consume significant bandwidth.

Therefore, the router should support Quality of Service (QoS).

QoS can prioritize:

- 1, Voice traffic.
- 2, Video conferencing.
- 3, Business-critical cloud applications.
- 4, Normal web browsing.
- 5, Non-critical downloads.

Cost and Scalability

The network can later be upgraded by:

- Increasing Internet bandwidth.
- Adding additional access points.
- Adding more switches.

Conclusion

A 4G/5G-based dual-WAN network with satellite/cellular backup, a managed router, firewall, switches, wireless access points and an encrypted VPN is an appropriate solution. It because balances bandwidth, reliability, security, cost and future scalability while allowing the rural office to maintain business operations.

Question 2:

University campus wireless network for 3000+ students

Problem Analysis

The university needs wireless Internet access for more than 3000 students in classrooms, laboratories, libraries and hostels.

The major constraints are:

- very large number of simultaneous users.
- Limited number of access points.
- High performance is required.

Proposed wireless Network Design

I recommend a centrally managed enterprise Wi-Fi network using Wi-Fi 6/6E-capable access points where supported.

The network should use:

Internet → Core Router/Firewall → Core Switches → PoE Switches → wireless Access points.

Access Point Placement

Since the number of access points is limited, they should not simply be installed at equal distances.

Access points should be placed based on user density and building structure.

Classrooms

APs should be installed near areas with high student density. Large lecture halls may require multiple access points because hundreds

of students can connect simultaneously.

Laboratories

Laboratories should have dedicated AP coverage because students may use laptops, tablets and other wireless devices simultaneously.

Libraries

Libraries require high-density wireless coverage because many students may connect simultaneously for long ~~time~~ periods.

Hostels

Hostels require APs distributed across floors and rooms/common areas according to the building layout.

Frequency Band Selection

The 5 GHz band should be preferred for most high-performance users because it

provides more non-overlapping channels and generally less interference than 2.4 GHz.

The 2.4 GHz band can be retained for devices that requires longer range or do not support 5 GHz.

Reducing Interference

To minimize interference:

- Use appropriate channel planning.
- Avoid excessive transmit power.
- Use automatic channel management.
- Prefer 5 GHz / 6 GHz for high-density areas.
- Perform periodic RF monitoring.
- Separate APs appropriately according to the site survey.

Handling 3000+ users

The network should support high-density client management.

Features Such as:

- Band steering
- Load balancing
- Airtime fairness
- Fast roaming
- Client limits
- QoS.

Authentication and Security

For a university environment,
WPA2-Enterprise or WPA3-Enterprise
with 802.1x/RADIUS authentication is
recommended.

Users can authenticate using their university
credentials.

Different wireless networks/VLANs can be
created for:

- students
- Faculty / staff

- Guests
- Network administrations
- IoT devices.

For example

Student Wi-Fi → Student VLAN

Faculty Wi-Fi → Faculty VLAN

Guest Wi-Fi → Guest VLAN

Balancing Coverage, Performance, security and Cost

This approach provides:

- Good coverage
- Better capacity
- Lower interference
- Better security.

Conclusion

A centrally managed high-density Wi-Fi network with strategically positioned Wi-Fi 6/6E access points 5 GHz/6 GHz as the preferred bands.

question 3:

Secure Network Architecture for a Financial Institution

Problem Analysis

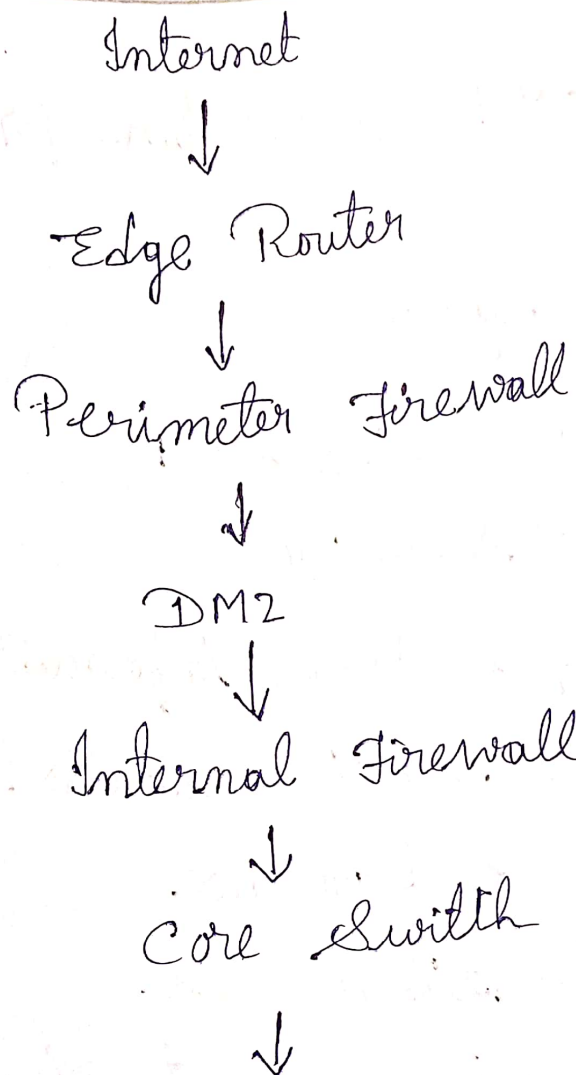
The main constraints are

- Strong cybersecurity
- Protection of customer information
- Secure online banking access.
- Good network performance.
- Controlled operational cost.
- Protection against unauthorized access and attacks.

Proposed Network Architecture

A defense-in-depth architecture should be used.

A simplified architecture is:



Segmented Internal VLANs

The network should not allow direct communication between every department.

1, Network Segmentation

Separate VLANs can be created for:

- employee systems.
- Customer - data Systems.

- Banking / application servers.
- Database servers.
- IT administration.
- Guest users.
- Security / monitoring systems.

2, DMZ

Public-facing services such as web servers should be placed in a DMZ (Demilitarized zone).

For example:

Internet → Firewall → DMZ web server

3, Firewall Deployment

A next-generation firewall should be placed at the network perimeter.

It should provide:

- Packet filtering
- Application control.

- VPN
- Intrusion prevention
- URL / content filtering
- Logging and monitoring
- Access - control policies.

4, Authentication

Recommended mechanisms include:

Multi-Factor Authentication (MFA)

Users can authenticate using:

- username / password
- One-time password / authenticator.
- Security token or another approved second factor

Role-based access control should also be implemented.

Users should receive only the permissions required for their jobs.

5, Intrusion Detection and Prevention

An IDS can detect:

- port scanning
- unusual traffic
- Malware-related activity
- Suspicious network behavior.

An IPS can automatically block or prevent identified malicious traffic.

Security logs should be centrally collected and monitored.

6, Protecting Sensitive Customer Data

sensitive customer information should be protected both:

- At rest using encryption.
- In transit using protocols such as TLS.

Regular backups should be maintained, and backups should be protected against unauthorized access.

7, Performance Considerations

Security should not unnecessarily slow down the network.

To maintain performance:

- use appropriately sized firewalls.
- Separate high-priority business traffic.
- use QoS where necessary.
- Monitor bandwidth and firewall utilization.
- Regularly review security rules.

critical banking traffic should receive appropriate priority.

Conclusion

A segmented, defense-in-depth network using VLANs, firewalls, DMZ, MFA/802.1X authentication, IDS/IPS encryption and centralized monitoring provides strong protection for the financial institution.